

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283661

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

SONG; Sang Heon et al.

Drying Apparatus for Secondary Battery

Abstract

A drying apparatus for secondary battery, wherein the drying apparatus dries an electrode sheet including an electrode current collector, a coated portion on which an electrode active material is applied on at least one surface of the electrode current collector, and a non-coated portion on which the electrode active material is not applied, wherein the coated portion is disposed to be parallel to a first direction to form at least one electrode line, includes an induction heating unit disposed to face the electrode sheet and drying the coated portion through electromagnetic induction, wherein the induction heating unit includes a multi-coil portion including a plurality of coils having different heating frequencies; and a switch portion provided to selectively apply current to any one of the plurality of coils.

Inventors: SONG; Sang Heon (Daejeon, KR), LEE; Dong Hyun (Daejeon, KR)

Applicant: SK On Co., Ltd. (Seoul, KR)

Family ID: 1000008389920

Appl. No.: 19/014417

Filed: January 09, 2025

Foreign Application Priority Data

KR 10-2024-0034024

Mar. 11, 2024

Publication Classification

Int. Cl.: F26B3/347 (20060101); H01M4/04 (20060101); H05B6/78 (20060101)

U.S. Cl.:

CPC F26B3/347 (20130101); H05B6/788 (20130101); H01M4/04 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent document claims the priority and benefits of Korean Patent Application No. 10-2024-0034024 filed on Mar. 11, 2024, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The disclosure and implementations disclosed in this patent document generally relate to a drying apparatus for secondary battery.

BACKGROUND

[0003] Secondary batteries, unlike primary batteries, have convenience of being able to be charged and discharged, and are thus receiving much attention as a power source for various mobile devices, electric vehicles, and the like.

[0004] Such secondary batteries may include battery cells in which an electrode assembly formed by stacking a positive electrode plate, a negative electrode plate, and a separator or by winding the same in a roll shape is accommodated in a case. A plurality of battery cells may be stacked in a predetermined direction and accommodated in a battery module or a battery pack.

[0005] Meanwhile, in a secondary battery manufacturing process, a positive electrode active material or a negative electrode active material may be thinly applied to a surface of a current collector of an electrode sheet, to form a positive electrode plate and a negative electrode plate, respectively. More specifically, the electrode sheet may include a non-coated portion, which is a portion of the current collector in which the electrode active material is not applied, and a coated portion, which is a portion of the current collector in which the electrode active material is applied.

[0006] Here, it is important to quickly and evenly dry the coated portion in terms of improving quality of the secondary battery and efficiency of the secondary battery manufacturing process.

SUMMARY

[0007] According to an aspect of the present disclosure, a drying apparatus for secondary battery, capable of quickly drying an electrode sheet, may be provided.

[0008] According to an aspect of the present disclosure, a drying apparatus for secondary battery, capable of improving drying quality, may be provided.

[0009] A battery cell used in a drying apparatus for secondary battery of the present disclosure may be widely applied in green technology fields such as an electric vehicle, a battery charging station or solar power generation, wind power generation, or the like using batteries, or the like. In addition, the battery cell of the present disclosure may be used in an eco-friendly electric vehicle, a hybrid vehicle, or the like to prevent a change in climate by suppressing air pollution and greenhouse gas emissions.

[0010] According to an embodiment of the present disclosure, provided is a drying apparatus for secondary battery, wherein the drying apparatus dries an electrode sheet including an electrode current collector, a coated portion on which an electrode active material is applied to at least one surface of the electrode current collector, and a non-coated portion on which the electrode active material is not applied, wherein the coated portion is disposed to be parallel to a first direction to form at least one electrode line, includes an induction heating unit disposed to face the electrode sheet and drying the coated portion through electromagnetic induction, wherein the induction heating unit includes a multi-coil portion including a plurality of coils having different heating frequencies; and a switch portion provided to selectively apply current to any one of the plurality of coils.

[0011] According to an embodiment, the switch portion may be connected to one of the plurality of coils having a heating frequency corresponding to the electrode sheet.

[0012] According to an embodiment, the plurality of coils may be provided to have different heating frequencies, respectively corresponding to a resonant frequency of at least one electrode sheet.

[0013] According to an embodiment, each of the plurality of coils may include at least one first heating coil portion facing the coated portion, and a plurality of second heating coil portions facing the non-coated portion, and the first heating coil portion and the second heating coil portion may be disposed in a second direction, perpendicular to the first direction.

[0014] According to an embodiment, the first heating coil portion and the second heating coil portion may be alternately disposed in the second direction.

[0015] According to an embodiment, a size of the second heating coil portion may be provided to be smaller than a size of the first heating coil portion, based on the first direction.

[0016] According to an embodiment, a size of the first heating coil portion in the second direction may be provided to correspond to a size of the coated portion in the second direction, and a size of the second heating coil portion in the second direction may be provided to correspond to a size of the non-coated portion in the second direction.

[0017] According to an embodiment, the electrode lines may be disposed in n rows (where n is a natural number other than 0), the plurality of coils may be configured to have different numbers of first heating coil portions, and the switch portion may be provided to selectively apply current to one coil having n first heating coil portions among the plurality of coils.

[0018] According to an embodiment, the plurality of coils may be disposed such that the number of the first heating coil portions increases in the first direction.

[0019] According to an embodiment, the multi-coil portion may be disposed as a plurality of multi-coil portions in the first direction.

[0020] According to an embodiment, the drying apparatus may further include a wavelength drying unit irradiating a wave having a predetermined frequency to dry the coated portion, and the wavelength drying unit may include at least one of a microwave drying portion irradiating a microwave to the electrode sheet or a laser drying portion irradiating a laser to the electrode sheet.

[0021] According to an embodiment, the induction heating unit and the wavelength drying unit may be disposed to face each other, with the electrode sheet interposed therebetween.

[0022] According to an embodiment, the wavelength drying unit may include both the microwave drying portion and the laser drying portion, and the microwave drying portion and the laser drying portion may be disposed alternately in the first direction.

[0023] According to an embodiment, the multi-coil portion may be disposed as a plurality of multi-coil portions in the first direction, and the plurality of multi-coil portions may be disposed to face one of the microwave drying portion or the laser drying portion.

[0024] According to an embodiment, at least one of the microwave drying portion or the laser drying portion and the multi-coil portion may be disposed alternately based on the first direction.

[0025] In addition, according to another embodiment of the present disclosure, a drying apparatus for secondary battery, wherein the drying apparatus dries an electrode sheet including an electrode current collector, a coated portion on which an electrode active material is applied on at least one surface of the electrode current collector, and a non-coated portion on which the electrode active material is not applied, wherein the coated portion forms n electrode lines disposed in n rows, parallel to a first direction, includes an induction heating unit disposed to face the electrode sheet and drying the coated portion through electromagnetic induction, wherein the induction heating unit includes a plurality of coils including a first heating coil portion corresponding to the coated portion to heat the coated portion, and a switch portion selectively applying current to any one of the plurality of coils, each of the plurality of coils has different numbers of first heating coil portions, and the switch portion selectively applies current to a coil having n first heating coil portions, among the plurality of coils (where n is a natural number other than 0).

[0026] The solutions according to the present disclosure have been described above, but are

illustrative, and it should be understood that other configurations, not mentioned, are included in the present disclosure even if they are added.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0027] Certain aspects, features, and advantages of the present disclosure may be illustrated by the following detailed description with reference to the accompanying drawings.

[0028] FIG. 1 is a schematic diagram of a drying apparatus for secondary battery according to an embodiment of the present disclosure.

[0029] FIG. 2 is a schematic diagram of a drying apparatus for secondary battery according to another embodiment of the present disclosure.

[0030] FIG. 3 is an exemplary diagram roughly illustrating an electrode sheet being dried through a drying apparatus for secondary battery of the present disclosure.

[0031] FIG. 4A is a view illustrating an electrode sheet of a first model in which coated portions are disposed in one row, and FIG. 4B is a view illustrating a multi-coil portion and a switch portion for drying an electrode sheet of a first model.

[0032] FIG. 5A is a view illustrating an electrode sheet of a second model in which coated portions are disposed in two rows, and FIG. 5B is a view illustrating a multi-coil portion and a switch portion for drying an electrode sheet of a second model.

[0033] FIG. 6A is a view illustrating an electrode sheet of a third model in which coated portions are disposed in three rows, and FIG. 6B is a view illustrating a multi-coil portion and a switch portion for drying an electrode sheet of a third model.

DETAILED DESCRIPTION

[0034] Before a detailed description of the embodiments, the terms or words used in the following description and claims should not be interpreted as limited to their usual or dictionary meanings, and should be interpreted as meanings and concepts that conform to the technical idea of the present disclosure based on the principle that the inventor may appropriately define the concept of the term in order to explain his own invention in the best way. The same reference numbers or symbols described in each drawing represent portions or components that perform substantially the same functions. For convenience of explanation and understanding, the same reference numbers or symbols may be used in different embodiments.

[0035] In the following description, singular expressions include plural expressions unless the context clearly indicates otherwise. It should be understood that terms such as “include,” “comprise,” or the like are intended to specify the presence of features, numbers, steps, operations, components, portions, or combinations thereof described in the specification, but do not preemptively exclude the possibility of the presence or addition of at least one other features, numbers, steps, operations, components, portions, or combinations thereof.

[0036] In addition, in the following description, expressions such as upward, above, on, upper portion, downward, below, lower portion, lateral, side surface, forward, front, rearward, rear, or the like may be expressed based on the direction illustrated in the drawings, and it should be noted in advance that when a direction of an object changes, it may be expressed differently.

[0037] Additionally, in this specification and claims, terms including ordinal numbers such as “first,” “second,” or the like may be used to distinguish between components. These ordinal numbers may be used to distinguish identical or similar components from each other, and the meaning of the term should not be interpreted limitedly due to the use of these ordinal numbers. For example, components combined with these ordinal numbers should not be interpreted as having a limited order of use or arrangement based on the number. As necessary, each ordinal number may be used interchangeably.

[0038] Hereinafter, the present disclosure will be described in detail with reference to the drawings. [0039] FIG. 1 is a schematic diagram of a drying apparatus for secondary battery according to an embodiment of the present disclosure, FIG. 2 is a schematic diagram of a drying apparatus 1 for secondary battery according to another embodiment of the present disclosure, and FIG. 3 is an exemplary diagram roughly illustrating an electrode sheet being dried through a drying apparatus 1 for secondary battery of the present disclosure.

[0040] Referring to FIGS. 1 to 3 together, a drying apparatus 1 for secondary battery of the present disclosure may include a roller unit 110 transporting an electrode sheet 10 in a predetermined direction, a coating unit 130 applying an electrode active material to the electrode sheet 10, and an induction heating unit 20 located to face the electrode sheet 10 and dry the electrode sheet 10.

[0041] Specifically, a drying apparatus 1 for secondary battery according to an embodiment of the present disclosure, as an electrode sheet 10 including an electrode current collector S, a coated portion 12 on which an electrode active material is applied on at least one surface of the electrode current collector S, and a non-coated portion 11 on which the electrode active material is not applied, in which the coated portion 12 may be disposed to be parallel to a first direction to form at least one electrode line L, may include an induction heating unit 20 disposed to face the electrode sheet 10 and drying the coated portion 12 through electromagnetic induction. In this case, the induction heating unit 20 may include a multi-coil portion 21 including a plurality of coils 211 having different heating frequencies, and a switch portion 25 provided to selectively apply current to any one of the plurality of coils 211 (see FIGS. 4A and 4B).

[0042] In addition, a wavelength drying unit 30 disposed to face the electrode sheet 10 to dry the electrode sheet 10, and a drying control unit 50 supplying and controlling power to the induction heating unit 20 and the wavelength drying unit 30, may be included.

[0043] The electrode sheet 10 may include an electrode current collector S and an electrode mixture layer E disposed on at least one surface of the electrode current collector S.

[0044] A portion of the electrode sheet 10 in which an electrode active material is applied and the electrode mixture layer E is disposed may be defined as the coated portion 12, and a portion of the electrode sheet 10 in which an electrode active material is not applied and the electrode mixture layer E is not disposed may be defined as the non-coated portion 11 (see FIGS. 4A, 5A, and 6A).

[0045] For example, the coated portion 12 may include the electrode current collector S and the electrode mixture layer E disposed on at least one surface of the electrode current collector S, but the non-coated portion 11 may include the electrode current collector S and may not include the electrode mixture layer E.

[0046] The electrode sheet 10 may form a positive electrode plate or a negative electrode plate, depending on a type of electrode active material applied to the electrode current collector S. For example, the electrode sheet 10 may include the positive electrode plate and the negative electrode plate. In this case, the positive electrode plate may be formed by arranging a positive electrode mixture layer including a positive electrode active material on a positive electrode current collector S among the electrode current collectors, and the negative electrode plate may be formed by arranging a negative electrode mixture layer including a negative electrode active material on a negative electrode current collector S among the electrode current collectors.

[0047] The electrode current collector S is not particularly limited as long as it has high electrical conductivity without causing a chemical change in battery. In addition, the electrode current collector S may be provided in the form of a metal thin film such as a film, a sheet, or a foil.

[0048] The electrode current collector S may include a positive electrode current collector and a negative electrode current collector.

[0049] The positive electrode current collector may include stainless steel, nickel, aluminum, titanium, or an alloy thereof. The positive electrode current collector may also include aluminum or stainless steel surface-treated with carbon, nickel, titanium, or silver. The positive electrode current collector may have a thickness of, but is not limited to, 10 to 50 μm , for example.

[0050] The negative electrode current collector may be, but is not limited to, a copper foil, a nickel foil, a stainless steel foil, a titanium foil, a nickel foam, a copper foam, a polymer substrate coated with a conductive metal, or the like. The negative electrode current collector may have a thickness of, but is not limited to, 10 to 50 μm , for example. The electrode composite layer E may be applied to at least one surface of the electrode current collector S.

[0051] The electrode composite layer E may include a positive electrode composite layer and a negative electrode composite layer.

[0052] The positive electrode composite layer may include a positive electrode active material. The positive electrode active material may include a compound capable of reversibly intercalating and deintercalating lithium ions. According to example embodiments, the positive electrode active material may include a lithium-nickel metal oxide. The lithium-nickel metal oxide may further include at least one of cobalt (Co), manganese (Mn), or aluminum (Al).

[0053] The negative electrode composite layer may include a negative electrode active material. A material capable of adsorbing and desorbing lithium ions may be used as the negative electrode active material. For example, as the negative electrode active material, a carbon-based material such as crystalline carbon, amorphous carbon, a carbon composite, or carbon fiber; lithium metal; lithium alloy; a silicon (Si)-containing material, a tin (Sn)-containing material, or the like may be used. The electrode sheet **10** may be wound around the roller unit **110**, and may be transported in an MD direction (lengthwise or Y-axis direction).

[0054] In a conventional hot air drying device, a surface of a coated portion may be directly heated, such that heat transfer may be performed in order of drying from an upper portion of the coated portion (far from a current collector) to a lower portion of the coated portion (close to the current collector). Therefore, a dry film may be formed on the surface of the coated portion. In this case, moisture in the coated portion (close to the current collector) may delay a drying time due to the dry film. In addition, when the dry film is generated, quality of an electrode assembly may deteriorate, such as defects due to the occurrence of cracks, boiling, or the like on the surface.

[0055] The present disclosure may minimize occurrence of the dry film, thereby securing quality of appearance of the electrode sheet **10** while shortening the drying time thereof. This will be described in detail below.

[0056] The present disclosure may include an induction heating unit **20** inductively heating the electrode sheet **10** using a plurality of coils **211** (see FIG. 4A).

[0057] The induction heating unit **20** may be disposed to face at least one of an upper surface or a lower surface of the electrode sheet **10**. In the drawings, the induction heating unit **20** is illustrated as being disposed below the electrode sheet **10**, but the present disclosure is not necessarily limited thereto.

[0058] The induction heating unit **20** may be disposed to face the electrode sheet **10**, and may heat the electrode sheet **10** through electromagnetic induction. More specifically, the induction heating unit **20** may include a plurality of coils **211**, and may heat the electrode current collector S of the electrode sheet **10** through induction heating.

[0059] According to an embodiment, the induction heating unit **20** of the present disclosure may heat the electrode current collector S, not the coated portion (**12**, specifically, the electrode mixture layer E applied to the coated portion **12**). The coated portion **12** may be dried by heating a lower surface (surface contacting the electrode current collector S) by the electrode current collector S. For example, heat may be transferred from a lower surface (close to the electrode current collector S) of the coated portion **12** to an upper surface (far from the electrode current collector; S) of the coated portion **12**.

[0060] Through this, the induction heating unit **20** of the present disclosure may minimize generation of a dry film and shorten a drying time by heating from an inside of the coated portion **12**.

[0061] Referring to FIGS. 4A to 6B in advance, the induction heating unit **20** may include a multi-

coil portion **21** including a plurality of coils **211**, and a switch portion **25** selectively applying current to the multi-coil portion **21**. The above-described structure will be described later in FIG. **4A** below.

[0062] According to an embodiment of the present disclosure, a wavelength drying unit **30** irradiating a wave having a predetermined frequency to dry the coated portion may be further included.

[0063] According to an embodiment, the wavelength drying unit **30** may include at least one of a microwave drying portion **31** irradiating a microwave to the electrode sheet **10** or a laser drying portion **33** irradiating a laser to the electrode sheet **10**.

[0064] As described above, the multi-coil portions **21** may be disposed in multiple numbers in the MD direction. In this case, according to an embodiment, the multi-coil portions **21** may be disposed to face at least one of the microwave portion **31** and the laser drying portions **33**.

[0065] In addition, according to an embodiment, the microwave drying portions **31** or the laser drying portions **33** may be disposed to be spaced apart from each other in a first direction (MD direction or Y-axis direction). In this case, the multi-coil portions **21** may be disposed to face each other in a height direction (Z-axis direction) in a space between the microwave portions **31** and the laser drying portions **33**.

[0066] For example, at least one of the microwave portions **31** or the laser drying portions **33** and the multi-coil portions **21** may be disposed alternately in the MD direction (first direction or Y-axis direction).

[0067] The wavelength drying unit **30** may be disposed to face at least one of an upper surface or a lower surface of the electrode sheet **10** to dry the electrode sheet **10**.

[0068] The wavelength drying unit **30** may include a microwave drying portion **31** applying a microwave having a frequency of 300 MHz or higher to the coated portion **12**, to generate heat in a dielectric (electrode mixture layer E of the coated portion **12**), and a laser drying portion **33** applying a laser beam having a wavelength of a predetermined range, to dry the electrode mixture layer E of the coated portion **12**.

[0069] The microwave drying portion **31** may generate heat in the electrode mixture layer E by using a frequency in the range of 300 MHz to 300 GHz. The microwave drying portion **31** may directly heat an interior of the electrode mixture layer E, which is a dielectric, without heating the electrode current collector S, which is a metal substrate. Through this, occurrence of the dry film may be minimized and a drying time may be reduced.

[0070] The laser drying portion **33** may dry the electrode mixture layer E by irradiating a laser beam to the electrode mixture layer E of the coated portion **12**. The laser drying portion **33** may irradiate the laser beam linearly or planarly to the upper surface (surface located far from the electrode current collector S) of the coated portion **12**.

[0071] The laser drying portion **33** may irradiate the laser beam having a wavelength of a predetermined range. For example, the laser drying portion **33** may irradiate the laser beam having a wavelength of 850 nm or less.

[0072] The range of this wavelength may be changed depending on purified water included in the undried electrode mixture layer E, and a material of the electrode mixture layer E.

[0073] For example, when the electrode mixture layer E is a negative active material including graphite, the laser drying portion **33** may irradiate the laser beam having a wavelength of 400 nm to 600 nm, more specifically, 600 nm or less. In this range, a large difference in light absorbance may be formed between purified water and graphite. Specifically, light absorbance of graphite may be greater than light absorbance of purified water, within this range.

[0074] Therefore, when irradiating the laser beam in the above-described range, graphite, which is the material of the electrode composite layer E, may be dried by heating and evaporating water. In addition, when an irradiation wavelength of the laser drying portion **33** exceeds 850 nm, absorption of purified water and absorption of graphite may gradually become similar to each other. In

addition, damage may be inflicted on the active material including graphite within this range. Therefore, a wavelength range irradiated by the laser drying portion **33** may be a wavelength range (850 nm or less) minimizing damage to the active material, but a specific wavelength range may be changed depending on a type of the active material. (For example, when the active material is graphite, the wavelength range may be 400 nm to 600 nm.)

[0075] The wavelength drying unit **30**) of the present disclosure may further include other drying means in addition to the microwave drying portion **31** and the laser drying portion **33**. For example, the wavelength drying unit **30** may further include general drying means such as an ultraviolet drying portion **35** irradiating a wavelength in an ultraviolet range, an air drying portion (not illustrated) supplying air for drying, or the like.

[0076] In addition, according to an embodiment, in the wavelength drying unit **30**, the microwave drying portion **31** and the laser drying portion **33** may be alternately disposed in the MD direction (Y-axis direction). However, the present disclosure is not limited to this arrangement.

[0077] In addition, the induction heating unit **20** and the wavelength drying unit **30**, described above, in the present disclosure may be disposed to face each other, with the electrode sheet **10** interposed therebetween. For example, the induction heating unit **20** may be disposed to face a first surface of the electrode sheet **10**, and the wavelength drying unit **30** may be disposed to face a second surface of the electrode sheet **10**.

[0078] For example, since the laser drying portion **33** irradiates a laser beam to the surface of the electrode mixture layer E, the laser drying portion **33** may be disposed to face the electrode mixture layer E. For example, the wavelength drying unit **30** and the electrode mixture layer E may face the electrode mixture layer E, without an obstacle (for example, an electrode current collector S) therebetween.

[0079] In addition, the induction heating unit **20** may heat the electrode current collector S, not the electrode mixture layer E, through electromagnetic induction, and, thus, may not be necessarily disposed to face the electrode mixture layer E.

[0080] Taking FIG. **3** as an example, when, in an electrode sheet **10** of the present disclosure, an electrode mixture layer E is applied to an upper surface of an electrode current collector S, a laser drying portion **33** may be disposed above the electrode sheet **10**, and, regardless thereof, an induction heating unit **20** may be disposed in at least one direction of an upward direction or a downward direction.

[0081] In addition, a drying apparatus **1** for secondary battery of the present disclosure may be disposed such that the induction heating unit **20** and the wavelength drying unit **30** face each other in the height direction (Z-axis direction), as illustrated in FIG. **1**. However, the present disclosure is not limited thereto, and the induction heating unit **20** and the wavelength drying unit **30** may be disposed alternately in a zigzag manner in the MD direction (Y-axis direction), as illustrated in FIG. **2**.

[0082] a drying apparatus **1** for secondary battery of the present disclosure may further include a drying control unit **50** controlling the induction heating unit **20** and the wavelength drying unit **30**.

[0083] The drying control unit **50** may appropriately control current, wavelength, or the like of the induction heating unit **20** and the wavelength drying unit **30**. In addition, the drying control unit **50** may include a power supply unit (not illustrated) that supplies power to the induction heating unit **20** and the wavelength drying unit **30**. In addition, the drying control unit **50** may also control selective current application of the switch portion **25** to be described later.

[0084] Hereinafter, a structure of the induction heating unit **20** of a drying apparatus **1** for secondary battery of the present disclosure will be described in detail with reference to FIGS. **4A** to **6B**. In the electrode sheet **10** of the present disclosure, the coated portion **12** may be applied lengthwise in the MD direction (Y-axis direction), to form electrode lines L disposed in at least one rows.

[0085] FIG. **4A** below illustrates an electrode sheet **10-1** ("first model") having one electrode line L

formed by the coated portion **12**, FIG. **5A** illustrates an electrode sheet **10-2** (“second model”) having two electrode lines **L**, and FIG. **6A** illustrates an electrode sheet **10-3** (“third model”) having three electrode lines **L**. However, the present disclosure is not limited thereto, and more electrode lines may be formed.

[0086] As described below, the present disclosure may be provided with an electrode sheet **10** having at least one electrode line **L**, and depending on the number of electrode lines **L**, one coil **211** may be selected from a multi-coil portion **21** by a switch portion **25**, described below.

[0087] FIG. **4A** is a view illustrating an electrode sheet of a first model in which coated portions are disposed in one row, and FIG. **4B** is a view illustrating a multi-coil portion and a switch portion for drying an electrode sheet of a first model.

[0088] Referring to FIGS. **4A** and **4B** together, an induction heating portion **20** of the present disclosure may include a multi-coil portion **21** including a plurality of coils **211**, and a switch portion **25** selectively connected to at least one of the plurality of coils **211**.

[0089] The multi-coil portion **21** may include a plurality of coils **211**. More specifically, the multi-coil portion **21** may include a plurality of coils **211** having different heating frequencies f . The plurality of coils **211** may have different shapes corresponding to a non-coated portion **11** and a coated portion **12** in an electrode sheet **10**. The plurality of coils **211** may include a first heating coil portion **211a** corresponding to the coated portion **12**, and a second heating coil portion **211b** corresponding to the non-coated portion **11**. The first heating coil portion **211a** and the second heating coil portion **211b** may be disposed in a direction (X-axis direction or second direction), perpendicular to an extension direction (Y-axis direction or first direction) of an electrode line **L**.

[0090] For example, the first heating coil portion **211a** and the second heating coil portion **211b** may be alternately disposed in the second direction (X-axis direction).

[0091] A term ‘corresponding (to)’ may mean being disposed to face each other. For example, the first heating coil portion **211a** may be disposed to face the coated portion **12**, and the second heating coil portion **211b** may be disposed to face the non-coated portion **11**. The plurality of coils **211** may be provided in different shapes, and may have different heating frequencies f . The heating frequency f may be a resonant frequency corresponding to one of the plurality of models of the electrode sheet **10**. The plurality of coils **211** may have different heating frequencies f and accordingly, may have different numbers of the first heating coil portions **211a**. For example, the plurality of coils **211** may be provided to have different heating frequencies f corresponding to resonant frequencies of at least one electrode sheets **10**.

[0092] For example, the plurality of coils **211** may include a first coil **211-1** having one first heating coil portion **211a**, a second coil **211-2** having two first heating coil portions **211a**, and a third coil **211-3** having three first heating coil portions **211a**.

[0093] In this case, the first coil **211-1** may generate a first heating frequency f_1 corresponding to an electrode sheet **10-1** of the first model, the second coil **211-2** may generate a second heating frequency f_2 corresponding to an electrode sheet **10-2** (see FIG. **5A**) of the second model, and the third coil **211-3** may generate a third heating frequency f_3 corresponding to an electrode sheet **10-3** (see FIG. **6A**) of the third model.

[0094] For example, among the plurality of coils **211**, an n th coil **211-n** may generate a heating frequency f corresponding to an electrode sheet **10-n** of an n th model, where the number of electrode lines **L** is n .

[0095] To concentrate induction heating on the coated portion **12** to which an electrode mixture layer **E** is applied, the induction heating unit **20** of the present disclosure may be provided with a shape in which a size (or width) of the second heating coil portion **211b** corresponding to the non-coated portion **11** in the MD direction (first direction or Y-axis direction) decreases. For example, the induction heating unit **20** may be provided with a shape in which a size decreases in a section corresponding to the non-coated portion **11**, an increase in temperature by induction heating in the non-coated portion **11** may be lower than that in the coated portion **12** due to the magnetic field

offset occurring in this portion.

[0096] Specifically, a size (or width) of the second heating coil portion **211b** may be provided to be smaller than a size (or width) of the first heating coil portion **211a**, based on the MD direction (first direction or Y-axis direction).

[0097] The switch portion **25** may be selectively electrically connected to one coil **211** among the plurality of coils **211**. The switch portion **25** may connect a coil **211** having a heating frequency f corresponding to a model of the electrode sheet **10**, i.e., the number of electrode lines L , and a power supply unit. For example, the switch portion **25** may be connected to any one coil among the plurality of coils **211** having a heating frequency f corresponding to the electrode sheet **10**.

[0098] Referring again to FIGS. **4A** and **4B**, an electrode sheet **10** may have a coated portion **12** disposed in one row, and an electrode line L may be disposed in one row. In this case, a switch portion **25** may supply power to a first coil **211-1** among a plurality of coils **211** having a heating frequency f_1 corresponding to a resonant frequency of a first model electrode sheet **10-1** in which the electrode lines L are disposed in one row.

[0099] For example, the present disclosure may perform electromagnetic induction heating optimized for the electrode sheet **10** by only controlling the switch portion **25** even when the number of electrode lines L is changed.

[0100] In a TD direction (second direction or X-axis direction) of the first model electrode sheet **10-1**, a width da_1 of the coated portion **12** may correspond to a width wa_1 of a first heating coil portion **211a** of a first coil **211-1**, and a width db_1 of the non-coated portion **11** may correspond to a width wb_1 of a second heating coil portion **211b** of the first coil **211-1**.

[0101] FIG. **5A** is a view illustrating an electrode sheet of a second model in which coated portions are disposed in two rows, and FIG. **5B** is a view illustrating a multi-coil portion and a switch portion for drying an electrode sheet of a second model.

[0102] Referring to FIGS. **5A** and **5B**, a second model electrode sheet **10-2** having two electrode lines L , and an operation of a switch portion **25** according thereto may be confirmed. The contents described above in FIGS. **4A** and **4B** will be omitted and described.

[0103] Referring to FIG. **5A**, an electrode sheet **10** may be a second model electrode sheet **10-2** in which a coated portion **12** is disposed in two rows, e.g., an electrode line L is disposed in two rows. Referring to FIG. **5B**, in this case, a switch portion **25** may supply power to a second coil **211-2** having a second heating frequency f_2 corresponding to a resonant frequency of the second model electrode sheet **10-2**, among a plurality of coils **211**.

[0104] In this case, a width wa_2 of a first heating coil portion **211a** may be provided to correspond to a width da_2 of a coated portion **12**, and a width wb_2 of a second heating coil portion **211b** may be provided to correspond to a width db_2 of a non-coated portion **11**.

[0105] Referring again to FIGS. **4A** to **5B**, a size (or width) of the first heating coil portion **211a** in the TD direction (second direction or X-axis direction) may be provided to correspond to a size (or width) of the coated portion **12** in the TD direction, and a size (or width) of the second heating coil portion **211b** in the TD direction may be provided to correspond to a size of the non-coated portion **11** in the TD direction.

[0106] FIG. **6A** is a view illustrating an electrode sheet of a third model in which coated portions are disposed in three rows, and FIG. **6B** is a view illustrating a multi-coil portion and a switch portion for drying an electrode sheet of a third model.

[0107] Referring to FIGS. **6A** and **6B**, a third model electrode sheet **10-3** having two electrode lines L , and an operation of a switch portion **25** according thereto may be confirmed. The contents described above in FIGS. **4A** and **4B** will be omitted and described.

[0108] Referring to FIG. **6A**, an electrode sheet **10** may be a third model electrode sheet **10-3** in which a coated portion **12** is disposed in three rows, e.g., an electrode line L is disposed in three rows. Referring to FIG. **6B**, in this case, a switch portion **25** may supply power to a second coil **211-2** having a third heating frequency f_3 corresponding to a resonant frequency of the third model

electrode sheet **10-3**, among a plurality of coils **211**.

[0109] In this case, a width **wa3** of a first heating coil portion **211a** may correspond to a width **da3** of a coated portion **12**, and a width **wb3** of a second heating coil portion **211b** may correspond to a width **db3** of a non-coated portion **11**.

[0110] As described above, a plurality of coils **211** may have different heating frequencies (**f1**, **f2**, **f3** . . .), depending on the number or a width (**wa1**, **wa2**, **wa3** . . .) of first heating coil portions **211a**.

[0111] As described above in FIGS. **4A** to **6B**, a plurality of electrode lines **L** may be formed.

[0112] For example, according to an embodiment of the present disclosure, in a drying apparatus for secondary battery for drying an electrode sheet **10** including an electrode current collector **S**, a coated portion **12** on which an electrode active material is applied on at least one surface of the electrode current collector **S**, and a non-coated portion **11** on which the electrode active material is be applied, wherein the coated portion **12** forms **n** electrode lines disposed in **n** rows, parallel to a first direction, the drying apparatus includes an induction heating unit **20** disposed to face the electrode sheet **10** and drying the coated portion **12** through electromagnetic induction, wherein the induction heating unit **20** includes a plurality of coils **211** including a first heating coil portion **211a** corresponding to the coated portion **12** to heat the coated portion **12**, and a switch portion **25** selectively applying current to any one of the plurality of coils **211**. In this case, each of the plurality of coils **211** may have different numbers of first heating coil portions **211a** (for example, the plurality of coils **211** may include a first coil **211-1** to an **n**th coil **211-n**), and the switch portion **25** may selectively applies current to a coil having **n** first heating coil portions, among the plurality of coils **211** (where **n** is a natural number other than 0).

[0113] Through this structure, the switch portion **25** may selectively supply current to one coil having an appropriate resonance frequency depending on a model type of the electrode sheet **10**, among the plurality of coils **211**, thereby performing optimal induction heating for a type of the electrode sheet **10** corresponding thereto.

[0114] For example, the switch portion **25** may supply current to a coil **211** having the number of first heating coil portions **211a** corresponding to the number of electrode lines **L** of the electrode sheet **10**, among the plurality of coils **211**. For example, according to an embodiment of the present disclosure, the electrode lines **L** may be disposed in **n** rows (where **n** is a natural number other than 0), and the plurality of coils **211** may be provided to have different numbers of first heating coil portions **211a**, but the switch portion **25** may be provided to selectively apply current to any one coil having **n** first heating coil portions **211a**, among the plurality of coils **211**.

[0115] For example, referring to FIGS. **6A** and **6B**, in an electrode sheet **10-3** having three electrode lines **L** (**n=3**), the switch portion **25** may supply current by switching to the third coil **211-3** having three first heating coil portions **211a**.

[0116] In addition, as described above, the first heating coil portion **211a** may be disposed between second heating coil portions **211b**. For example, the plurality of coils **211** may include a first coil **211-1** including one first heating coil portion **211a** and two second heating coil portions **211b**, a second coil **211-2** including two first heating coil portions **211a** and three second heating coil portions **211b**, and a third coil **211-3** including three first heating coil portions **211a** and four second heating coil portions **211b**.

[0117] In the above embodiment, the first coil **211-1** to the third coil **211-3** may be disposed such that the number of the first heating coil portions **211a** sequentially increases in the TD direction. The number of the first heating coil portions **211a** may increase from the first coil **211-1** to the third coil **211-3**. In this manner, the plurality of coils **211** according to an embodiment may be disposed such that the number of the first heating coil portions **211a** increases in the MD direction (first direction or the Y-axis direction).

[0118] For example, as illustrated in the drawings, the first heating coil portions **211a** may be disposed such that the number thereof increases in a direction (Y-axis direction) from the first coil

211-1 to the third coil **211-3**.

[0119] Referring to FIGS. **4B**, **5B**, and **6B**, a plurality of coils **211** in a multi-coil portion **21** may be disposed such that the number of the first heating coil portions **211a** increases in the MD direction (Y-axis direction).

[0120] In addition, the multi-coil portions **21** respectively having the plurality of coils **211** may be disposed in plural in the MD direction (first direction or the Y-axis direction).

[0121] Although various embodiments the present disclosure have been described in detail above, the scope of the present disclosure is not limited thereto, and it will be obvious to those with average knowledge in the relevant technical field that various modifications and variations may be possible within a scope that does not depart from the technical idea of the present disclosure described in the claims. Some components of the above-described embodiments may be deleted and implemented, and each embodiment may be implemented in combination with each other. The contents described above may be merely examples of applying the principles of the present disclosure, and other configurations may be further included within a scope that does not depart from the scope of the present disclosure.

[0122] A drying apparatus for secondary battery according to an embodiment of the present disclosure may quickly dry an electrode sheet.

[0123] A drying apparatus for secondary battery according to an embodiment of the present disclosure may improve drying quality of an electrode sheet.

[0124] Only specific examples of implementations of certain embodiments may be described. Variations, improvements and enhancements of the disclosed embodiments and other embodiments may be made based on the disclosure of this patent document.

Claims

1. A drying apparatus for secondary battery, wherein the drying apparatus dries an electrode sheet including an electrode current collector, a coated portion on which an electrode active material is applied on at least one surface of the electrode current collector, and a non-coated portion on which the electrode active material is not applied, wherein the coated portion is disposed to be parallel to a first direction to form at least one electrode line, comprising: an induction heating unit disposed to face the electrode sheet and drying the coated portion through electromagnetic induction, wherein the induction heating unit includes: a multi-coil portion including a plurality of coils having different heating frequencies; and a switch portion provided to selectively apply current to any one of the plurality of coils.
2. The drying apparatus of claim 1, wherein the switch portion is connected to one of the plurality of coils having a heating frequency corresponding to the electrode sheet.
3. The drying apparatus of claim 1, wherein the plurality of coils are provided to have different heating frequencies, respectively corresponding to a resonant frequency of at least one electrode sheets.
4. The drying apparatus of claim 1, wherein each of the plurality of coils includes at least one first heating coil portion facing the coated portion, and a plurality of second heating coil portions facing the non-coated portion, and wherein the first heating coil portion and the second heating coil portions are disposed in a second direction, perpendicular to the first direction.
5. The drying apparatus of claim 4, wherein the first heating coil portion and the second heating coil portion are disposed alternately in the second direction.
6. The drying apparatus of claim 4, wherein a size of the second heating coil portion is smaller than a size of the first heating coil portion, based on the first direction.
7. The drying apparatus of claim 4, wherein a size of the first heating coil portion in the second direction is provided to correspond to a size of the coated portion in the second direction, and a size of the second heating coil portion in the second direction is provided to correspond to a size of the

non-coated portion in the second direction.

8. The drying apparatus of claim 4, wherein the electrode lines are disposed in n rows (where “n” is a natural number other than 0), the plurality of coils are configured to have different numbers of first heating coil portions, and the switch portion is provided to selectively apply current to one coil having n first heating coil portions among the plurality of coils.

9. The drying apparatus of claim 8, wherein the plurality of coils are disposed such that the number of first heating coil portions increases in the first direction.

10. The drying apparatus of claim 1, wherein the multi-coil portion is disposed as a plurality of multi-coil portions in the first direction.

11. The drying apparatus of claim 1, further comprising a wavelength drying unit irradiating a wave having a predetermined frequency to dry the coated portion, and wherein the wavelength drying unit includes at least one of a microwave drying portion irradiating a microwave to the electrode sheet or a laser drying portion irradiating a laser to the electrode sheet.

12. The drying apparatus of claim 11, wherein the induction heating unit and the wavelength drying unit are disposed to face each other, with the electrode sheet interposed therebetween.

13. The drying apparatus of claim 12, wherein the wavelength drying unit includes both the microwave drying portion and the laser drying portion, and the microwave drying portion and the laser drying portion are disposed alternately in the first direction.

14. The drying apparatus of claim 13, wherein the multi-coil portion is disposed as a plurality of multi-coil portions in the first direction, and the plurality of multi-coil portions are respectively disposed to face one of the microwave drying portion or the laser drying portion.

15. The drying apparatus of claim 11, wherein at least one of the microwave drying portion or the laser drying portion and the multi-coil portion are disposed alternately based on the first direction.

16. A drying apparatus for secondary battery, wherein the drying apparatus dries an electrode sheet including an electrode current collector, a coated portion on which an electrode active material is applied on at least one surface of the electrode current collector, and a non-coated portion on which the electrode active material is not applied, wherein the coated portion forms n electrode lines disposed in n rows, parallel to a first direction, comprising: an induction heating unit disposed to face the electrode sheet and drying the coated portion through electromagnetic induction, wherein the induction heating unit includes a plurality of coils including a first heating coil portion corresponding to the coated portion to heat the coated portion, and a switch portion selectively applying current to any one of the plurality of coils, each of the plurality of coils has different numbers of first heating coil portions, and the switch portion selectively applies current to a coil having n first heating coil portions, among the plurality of coils (where “n” is a natural number other than 0).
